ITCS-5122

The Economic Recovery *Trackers*

Team 10

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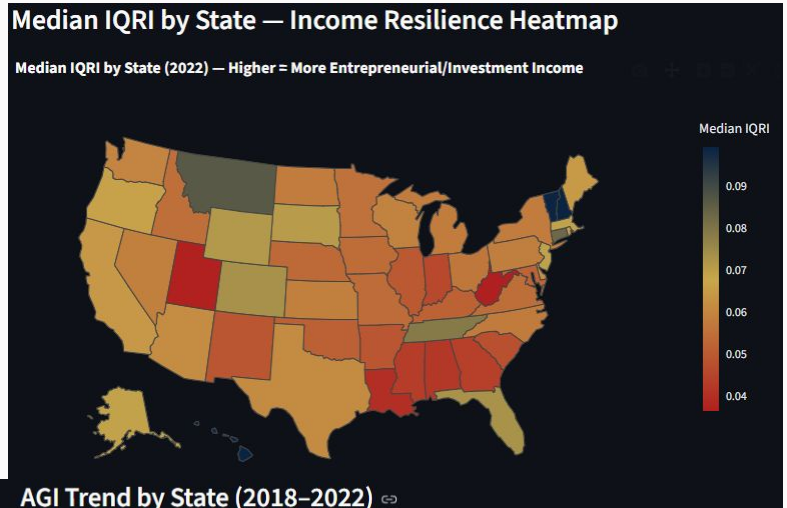
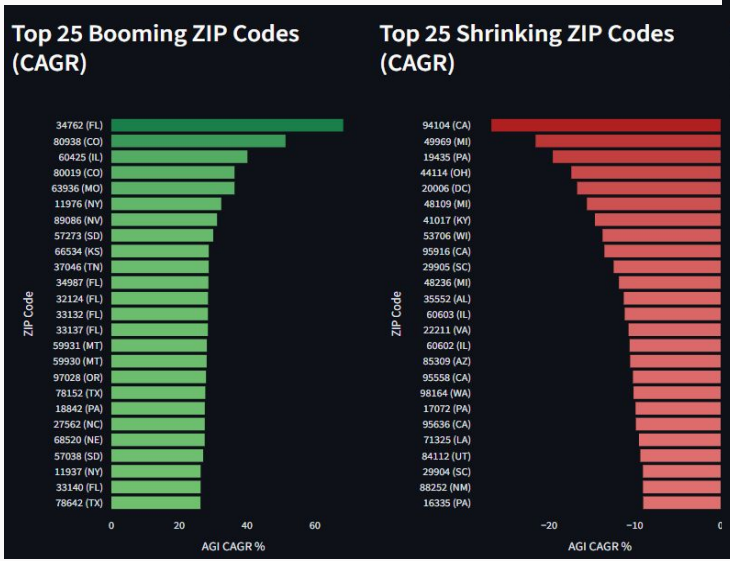
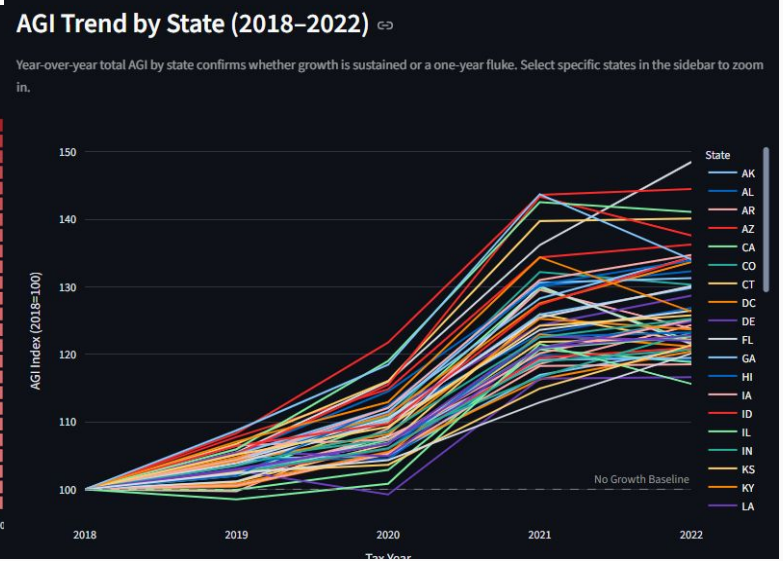
The **MVP** *Demo*

Primary Task: Compare historical tax returns against current ones, visualize which ZIP codes are seeing an exodus of wealth versus those that are booming.







Visual Grammar Defense

Bar Charts *- used for ranking and comparisons*

Channel: Length (AGI CAGR %)

Encoding: Sorted horizontal layout

Why this works: *Enables quick comparison and clear ranking*

Scatter Plot *- used for relationship analysis*

Marks: Points (ZIP codes)

Channels:

X → AGI of Baseline Year

Y → AGI of Latest Year

Color → CAGR (diverging scale)

Enhancements:

Log scale (wide income range)

Reference line (y = x)

Why this works: *Reveals relationships and highlights growth patterns*

Line Chart *- use to chart temporal trends*

Marks: Lines

Channel: Position over time

Why this works: *Effectively shows changes over time.*

Choropleth Map *- used to chart Geographic Pattern*

Marks: Regions (states)

Channel: Color (median CAGR)

Why this works: *Illustrate economic growth/decline across geographic areas*

Donut Chart *- used to show income tier distribution*

Marks: Segments (income tier)

Channel: Angle (proportion of each tier), Color (income tier category)

Why this works: *Compares the relative proportions of each income tier*

The "Phase 2" Demo

Income Quality & Resilience Index (IQRI) = (Business Income [A00900] + Dividends [A00600]) / Wages [A00200]

Tier	IQRI Range	Lending Implication
Wage-Dependent	< 0.05	High layoff risk, conservative lending only
Mixed	0.05 - 0.15	Balanced income, standard underwriting
Entrepreneurial	0.15 - 0.30	Strong business ownership, commercial lending targets
Investment-Led	0.30 >	Prime wealth management prospects

https://drive.google.com/file/d/1H3183N2yjEsVM9VycAyqi_Jpzx92nM-p/view?usp=sharing

AI Integration (Defined)

The Role of AI in the Project:

- Interpret user intent
- Recommend visualizations
- Generate insights

The pipeline we have planned:

User query (natural language)

Natural language query

Intent classification

Extract:

- *Metrics → CAGR, IQRI*
- *Constraints → high, low, top N*
- *Entities → ZIP, state*

Feature selection

Map intent → dataset columns

Visualization recommendation

Rule-based / ML-assisted mapping:

- *Comparison → Bar Chart*
- *Relationship → Scatter Plot*
- *Trend → Line Chart*

Output

Auto-generated visualization

Insight summary

Sprint to Final

AI implementation

- *Rule based parser*
- *Feature Mapping*
- *Visualization recommendation engine*

System Integration

- *Connect AI layer to Streamlit UI*
- *Enable Query box*
- *Enable Dynamic Chart Generation*

UI/UX Refinement

- *Improve layout consistency*
- *Add Tooltips & metric explanations*
- *Add Clear labeling (IQRI tiers, CAGR meaning)*
- *Optimize responsiveness + load feedback*

Validation & Testing

- *Verify Correct filtering logic*
- *Verify Accurate metric calculations (CAGR, IQRI)*
- *Test edge cases:*
  - *Missing data*
  - *Low-population ZIPs*

Thank you